

Business Requirements Document

Vendor Concentration Risk & Slow-Moving Inventory Reduction Initiative

Document Type:	Business Requirements Document (BRD)
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Source Analysis:	Vendor Performance Analysis (SQL / Python, 2026)

1. Executive Summary

An analysis of vendor purchasing, sales, and inventory data uncovered two related business problems that jointly expose the organization to financial and operational risk: a high concentration of procurement spend in a small number of vendors, and a significant amount of capital tied up in unsold inventory. This document defines the business requirements for an initiative to address both problems together, since they share the same root causes (purchasing behavior and vendor management practices) and the same stakeholders.

65.7%

of total purchase spend flows through just 10 of 119 vendors

\$2.71M

in unsold inventory currently locked up as working capital

~72%

lower unit cost achievable through consolidated bulk ordering

This initiative recommends: (1) formal supplier concentration risk monitoring and diversification planning, (2) a structured slow-moving inventory clearance process, and (3) updated purchasing policy to capture available bulk-order savings. Expected outcomes include reduced single-vendor dependency risk, recovery of a meaningful share of the \$2.71M currently locked in unsold stock, and measurable unit-cost savings on future purchase orders.

2. Business Problem & Objective

2.1 Problem Statement

The organization currently has no formal process for monitoring vendor concentration risk or for systematically identifying and clearing slow-moving inventory. As a result, purchasing decisions are made vendor-by-vendor and order-by-order, without visibility into (a) how dependent the business is on its largest suppliers, or (b) how much capital is sitting idle in unsold stock at any given time. This creates exposure to supply disruption and unnecessary carrying costs, and leaves cost-saving opportunities (like bulk-order pricing) unrealized.

2.2 Business Objectives

ID	Objective	Success Measure
OBJ-1	Reduce reliance on any single vendor or small vendor group for critical purchase volume	Top-10 vendor purchase concentration reduced from 65.7% to a target ceiling (e.g. ≤55%) within 12 months
OBJ-2	Reduce capital tied up in unsold inventory	Unsold inventory value reduced from \$2.71M by an agreed target percentage within 2 quarters
OBJ-3	Capture available bulk-purchasing cost savings	Increased share of purchase orders placed at "Large" order-size tier; measurable average unit-cost reduction
OBJ-4	Establish ongoing visibility into vendor and inventory risk	Recurring (e.g. quarterly) vendor concentration and inventory turnover reporting in place

2.3 Scope

In scope:

- Vendors and brands covered by the existing vendor sales/purchase summary dataset
- Purchasing policy and process changes related to order consolidation and bulk-order tiers
- A defined process for identifying and acting on slow-moving/unsold inventory
- Recurring reporting on vendor concentration and inventory turnover

Out of scope:

- Vendor contract renegotiation execution (handled by Procurement/Legal outside this initiative)
- IT/system replacement – this initiative assumes the existing inventory database and reporting tooling are used
- New product line or category decisions

3. Stakeholder Analysis

Stakeholder	Role	Interest / Stake	Influence / Involvement
VP Procurement	Executive Sponsor	Owns vendor relationships and supplier risk; accountable for procurement cost and continuity	High – approves vendor diversification & policy changes
Finance / FP&A	Key Stakeholder	Owns working-capital and cash-flow impact of unsold inventory	High – approves clearance/write-down decisions and budget impact
Category Managers	Process Owner	Manage day-to-day vendor and product-level	High – implement new purchasing

		purchasing decisions	thresholds and clearance actions
Warehouse / Inventory Operations	Process Owner	Physically manage stock; execute clearance and returns processes	Medium – operational execution of To-Be process
Data/BI Analyst	Support	Maintains the underlying vendor performance data and reporting	Medium – builds recurring reporting requirement
Individual Top-10 Vendors	External	Revenue relationship with the business; may be approached re: bulk terms or diversification conversations	Low (informed, not decision-making) – but high dependency in the other direction

4. Current State Findings (Data-Driven)

The following findings come from analysis of vendor purchase, sales, and invoice data, and form the business justification for this initiative.

4.1 Vendor Concentration

- Of 119 total vendors, the top 10 account for **65.7%** of total purchase dollars.
- The four largest vendors alone (Diageo North America, Martignetti Companies, Pernod Ricard USA, Jim Beam Brands) represent a substantial share of that concentration.
- No current process exists to monitor this ratio over time or flag it as a risk threshold is approached.

4.2 Slow-Moving / Unsold Inventory

- Total unsold inventory value across all vendors: **\$2.71M**.
- This is concentrated in a small number of vendors – Diageo (\$722K), Jim Beam (\$555K), Pernod Ricard (\$471K), and William Grant & Sons (\$402K) account for the largest shares.
- A defined group of vendors show stock turnover below 1.0 (purchases outpacing sales), indicating ongoing over-purchasing relative to demand.

4.3 Purchasing Behavior

- Average unit purchase price drops from ~\$39.07 (small orders) to ~\$10.78 (large orders) — a ~72% reduction — but current purchasing does not appear to systematically target the large-order tier.
- A specific set of brands combine low sales volume with high profit margin, indicating under-utilized promotional opportunity rather than a pricing problem.

Why this matters financially: \$2.71M in unsold inventory is capital that could otherwise fund operations, debt reduction, or new purchasing — and it continues to accrue holding costs the longer it remains unsold. Combined with high vendor concentration, the business carries both a liquidity risk and a supply continuity risk simultaneously.

5. Process: As-Is vs. To-Be

5.1 As-Is Process — Slow-Moving Inventory (Current State)

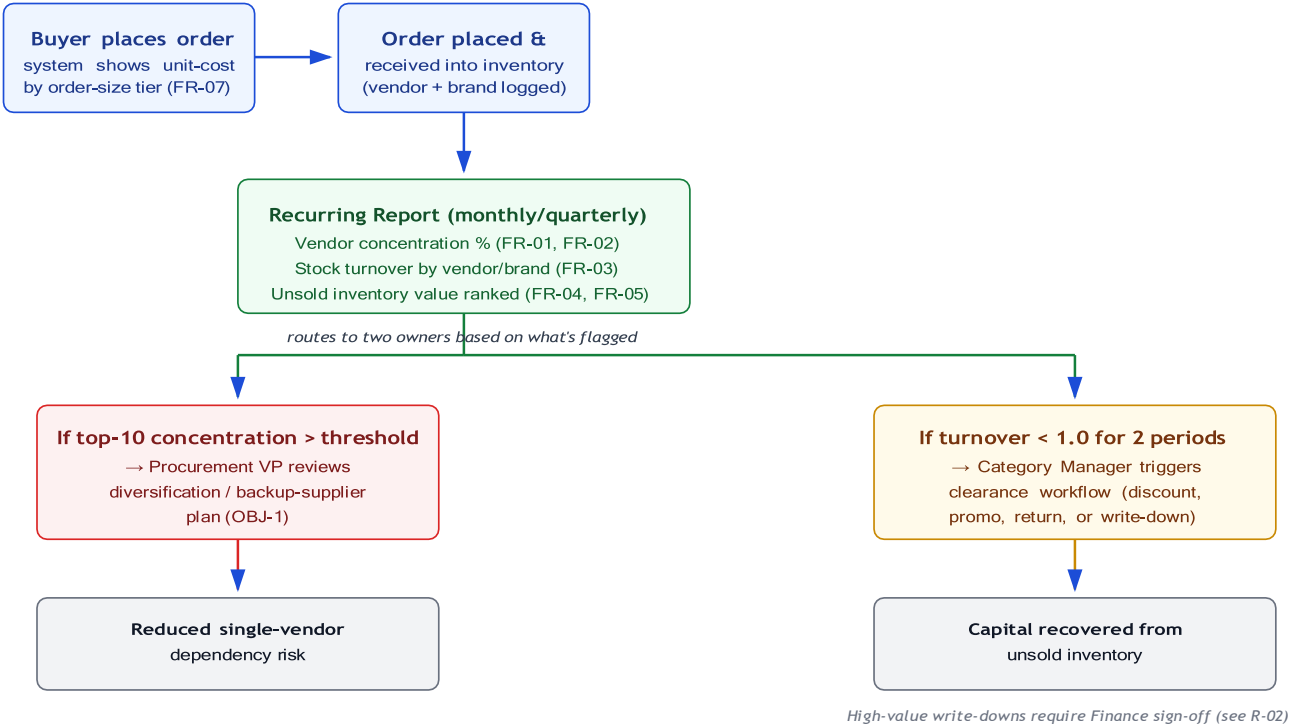
Today, there is no formal trigger for identifying or acting on slow-moving inventory. Vendor purchasing decisions are made independently at the category level, without a shared view of vendor concentration or turnover across the business.



*No recurring vendor concentration check happens anywhere in this flow.
Purchasing decisions are made per-vendor, per-category, without a shared risk view.*

5.2 To-Be Process — Vendor & Inventory Risk Management (Proposed State)

The proposed process introduces a recurring review cycle that surfaces both concentration risk and slow-moving inventory automatically, and routes each to the right owner for action.



5.3 Key Process Gaps Addressed

Gap in As-Is	Addressed in To-Be
No visibility into vendor concentration until a problem occurs	Recurring automated concentration report with defined risk thresholds
Slow-moving inventory identified reactively, if at all	Recurring turnover report flags vendors/SKUs below turnover threshold on a schedule
No standard action when excess inventory is found	Defined clearance workflow (discount, promotion, return-to-vendor, or write-down) with clear decision owner
Purchasing decisions don't factor in bulk-order pricing tiers	Purchasing policy updated to prefer consolidated/large-tier orders where demand allows

6. Functional Requirements

Req ID	Requirement	Priority	Owner
FR-01	System/report shall calculate and display total purchase dollars by vendor, and each vendor's % share of total spend, on a recurring (at least quarterly) basis	Must Have	Data/BI Analyst
FR-02	Report shall flag when the top-10 vendor concentration exceeds an agreed threshold (e.g. 60%)	Must Have	Data/BI Analyst
FR-03	System shall calculate stock turnover ratio (sales quantity / purchase quantity) by vendor and by brand	Must Have	Data/BI Analyst
FR-04	System shall calculate unsold inventory value per vendor and per brand, updated on a recurring basis	Must Have	Data/BI Analyst
FR-05	Report shall rank and surface the vendors/brands contributing the most to unsold inventory value	Should Have	Data/BI Analyst
FR-06	Process shall define an escalation path when a vendor's turnover ratio falls below 1.0 for two consecutive review periods	Should Have	Category Manager
FR-07	Purchasing workflow shall present unit-price impact of order size (Small/Medium/Large tier) at time of order placement	Should Have	Procurement / IT
FR-08	Report shall identify brands with sales in the bottom 15th percentile but profit margin in the top 15th percentile, for promotional review	Could Have	Data/BI Analyst
FR-09	System shall maintain a historical log of concentration % and unsold inventory value over time, to track trend against targets	Could Have	Data/BI Analyst

6.1 Non-Functional Requirements

Req ID	Requirement	Rationale
NFR-01	Vendor concentration and inventory reports must refresh at least quarterly, ideally monthly	Risk conditions can change between long reporting cycles
NFR-02	Reports must be accessible to non-technical stakeholders (Procurement, Finance) without requiring SQL/Python knowledge	Primary audience is business, not technical, users
NFR-03	Underlying data (vendor_sales_summary table) must remain auditable back to source purchase/sales records	Supports Finance sign-off on any write-down or clearance decision

7. User Stories & Acceptance Criteria

US-01

As a Procurement leader, I want to see each vendor's share of total purchase spend, so that I can identify when concentration risk is building up.

Acceptance Criteria:

- Given the vendor summary data, when the report is generated, then each vendor's purchase \$ and % of total spend is displayed, sorted descending.
- Given the top-10 vendor group, when their combined share exceeds the defined threshold, then the report visually flags this (e.g. red indicator).

US-02

As a Category Manager, I want to see which vendors/brands have the most unsold inventory value, so that I can prioritize clearance actions where they'll free up the most capital.

Acceptance Criteria:

- Given vendor-level unsold inventory values, when the report is run, then vendors are ranked descending by unsold inventory \$.
- Given a vendor with stock turnover below 1.0, when viewing that vendor's detail, then the specific brands/SKUs driving the low turnover are listed.

US-03

As a Buyer, I want to see the unit-price impact of order size before submitting a purchase order, so that I can choose the most cost-effective order quantity.

Acceptance Criteria:

- Given a draft purchase order, when quantity is entered, then the estimated unit price for that order-size tier (Small/Medium/Large) is displayed.
- Given an order that could reach a lower-cost tier with a small quantity increase, when in the Small or Medium tier, then the system suggests the potential savings.

US-04

As a Finance analyst, I **want to** track unsold inventory value and vendor concentration over time, **so that** I can report progress against this initiative's targets to leadership.

Acceptance Criteria:

- Given historical report snapshots, when viewing the trend report, then unsold inventory \$ and top-10 concentration % are shown period-over-period.
- Given a target (e.g. reduce unsold inventory by 20% in 2 quarters), when the trend report is viewed, then progress toward that target is visible.

8. Risks, Assumptions & Constraints

8.1 Risks

ID	Risk	Impact	Mitigation
R-01	Diversifying away from top vendors could raise unit costs if alternative suppliers lack the same bulk-pricing leverage	High	Diversify gradually; renegotiate volume tiers with current top vendors in parallel rather than abrupt shift
R-02	Aggressive inventory clearance (discounting) could erode margin in the short term	Medium	Set clearance discount limits; involve Finance in approval for high-value write-downs
R-03	Category Managers may resist standardized purchasing thresholds that reduce local flexibility	Medium	Involve Category Managers in threshold-setting; frame as guidance, not hard restriction, initially
R-04	Underlying data quality issues (e.g. negative gross profit rows previously excluded from analysis) could understate the true problem	Low	Periodic data quality review as part of the recurring reporting process

8.2 Assumptions

- The existing vendor_sales_summary dataset and its refresh process remain available and reasonably current.
- Procurement and Finance leadership are willing to define and agree on concentration and turnover thresholds.
- Vendors are open to bulk-order or volume-tier conversations without requiring full contract renegotiation cycles.

8.3 Constraints

- This initiative does not include budget for new software/BI tooling; it assumes existing reporting infrastructure is extended.
- Any vendor contract changes are subject to standard Procurement/Legal review timelines outside this BRD's control.

9. Expected Business Impact

Area	Current State	Target / Expected Impact
Vendor concentration	65.7% of spend in top 10 vendors	Reduced concentration and documented contingency plan for top vendors within 12 months
Unsold inventory / working capital	\$2.71M locked in unsold stock	Meaningful reduction (target to be set with Finance) within 2 quarters via structured clearance
Purchase unit cost	Order-size tiers not systematically used	Increased share of large-tier orders; measurable average unit-cost reduction
Risk visibility	No recurring reporting	Quarterly (or more frequent) vendor concentration & inventory turnover reporting in place

This BRD is derived from the Vendor Performance Analysis dataset and notebook analysis (vendor_performance_analysis.ipynb / eda.ipynb). All figures reflect the source dataset as analyzed and should be revalidated against current data before formal sign-off.